

FINAL PROJECT REPORT

Food Ordering Behaviour Analysis

Data Analytics & Visualization using Tableau

Student	Mehak Chugh
Project Area	Food Ordering Behaviour
Technology	Tableau
Report Date	31 July 2026
Project Period	17-30 July 2026

This report follows the supplied Final Project Report Template and covers planning, data preparation, visualization, dashboard/story development, performance testing, observations, future scope and appendix.

1. Introduction

1.1. Project Overview

The Food Ordering Behaviour Analysis project uses Tableau to transform food-ordering data into interactive business insights. The project examines how customer characteristics, meal type, cuisine, mood, hunger level, weather, day type, discounts, ratings, delivery fee and ordering time are associated with customer spending and ordering behaviour. The objective is to move from raw data to clear, decision-oriented visual analysis through a structured data analytics workflow.

1.2. Objectives

- Identify important patterns in customer ordering behaviour and spending.
- Compare order value across cities, cuisines, meal types, time periods and customer segments.
- Study the relationship between ratings and repeat-order behaviour.
- Understand the influence of weather, discounts, mood and hunger level on customer spending.
- Analyse delivery fee and time taken to order across meal type and day type.
- Build interactive Tableau visualizations and a consolidated dashboard/story for presentation.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Food-ordering platforms generate large amounts of customer and order information, but raw records alone do not clearly show which factors drive spending and ordering patterns. The project therefore addresses the problem of converting food-ordering data into meaningful visual insights that can support understanding of customer preferences, order value and operational behaviour.

2.2. Project Proposal (Proposed Solution)

The proposed solution is a Tableau-based analytical dashboard and story. The workflow includes data collection, quality assessment, preprocessing, business-question framing, calculated fields, visualization development and dashboard integration. Interactive visuals allow the analysis to be explored from multiple dimensions rather than relying on a single summary.

2.3. Initial Project Planning

Activity	Planned Period
Data collection and preparation	17-19 July 2026
Data visualization	20-23 July 2026
Dashboard development	25-26 July 2026
Presentation preparation	29-30 July 2026
Report date	31 July 2026

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The project used a food-ordering dataset containing order and customer-related attributes required for behavioural analysis. The analytical fields considered include city, cuisine, meal type, time of day, weather, customer rating, age, hunger level, restaurant type, dining company, mood, discount offers, delivery fee, day type, repeat-order behaviour and time taken to order.

3.2. Data Quality Report

- The raw dataset was reviewed for completeness, consistency and suitability for Tableau analysis.
- Fields were checked for meaningful values and compatibility with the intended business questions.
- Categorical fields were standardized where required so that categories could be compared consistently.
- Numerical fields such as order value, customer rating, delivery fee and ordering time were prepared for aggregation and comparison.

3.3. Data Exploration and Preprocessing

Exploratory analysis was used to understand the distribution and structure of the available variables before visualization. Calculated fields and grouping logic were used where necessary to create analytical categories, including customer age groups and rating categories. The prepared data was then connected to Tableau for visual analysis.

4. Data Visualization

4.1. Framing Business Questions

The project was organized around 11 business questions:

- 1. Which cities generate the highest order value?
- 2. Which cuisine receives the highest number of customer ratings across different meal types?
- 3. How does rainy weather influence order value across different times of the day?
- 4. How do customer ratings influence repeat orders among different age groups?
- 5. Does hunger level affect spending on different cuisines?
- 6. How do customer rating categories vary across different restaurant types based on the customer's dining company?
- 7. Which time of day generates the highest order value for each hunger level?
- 8. How do average customer ratings vary across different age groups and cities?
- 9. How do rainy weather, discount offers, and customer mood influence customer spendings?
- 10. Analysis of meal type and day type simultaneously on the basis of delivery fee and time taken to order.
- 11. How does meal type ordered vary with variation in mood?

4.2. Developing Visualizations

The project developed Tableau visuals selected according to the analytical requirement of each business question. The visualization set included map, heat-map, side-by-side bar, stacked bar, horizontal bar, packed bubble, pie and radial-style analyses. Nine core visuals were used in the final dashboard, while the broader business-question set supported the accompanying story/report analysis.

#	Business Question / Analysis	Visualization Approach
1	Highest order value by city	Map chart
2	Cuisine ratings across meal types	Heat chart
3	Rainy weather and order value by time	Side-by-side bar chart
4	Ratings and repeat orders by age group	Stacked bar chart
5	Hunger level and spending by cuisine	Horizontal bar chart
6	Rating categories by restaurant type and dining company	Packed bubble chart
7	Order value by time of day and hunger level	Pie chart
8	Average rating by age group and city	Heat map
9	Weather, discount and mood versus spending	Radial chart

5. Dashboard

5.1. Dashboard Design File

The Tableau dashboard consolidates the major visuals into an interactive analytical view. The design focuses on comparing customer behaviour across demographic, contextual and order-related dimensions. The dashboard was designed to make important patterns easy to identify while allowing the user to explore categories through Tableau interactions and filters.

- Core dashboard visuals: 9.
- Interactive analysis through category-based comparisons and filters.
- Primary analytical themes: order value, ratings, repeat orders, cuisine, meal type, weather, mood and customer behaviour.

Tableau Dashboard

CUSTOMER_ORDERING_ANALYSIS / Dashboard1

https://public.tableau.com/app/profile/mehak.chugh/viz/CUSTOMER_ORDERING_ANALYSIS/Dashboard1

6. Report

6.1. Story Design File

The Tableau story provides a sequential presentation of the analysis, connecting individual business questions with their corresponding visual evidence. It supports a clear demonstration of how the project moves from customer/order characteristics to broader behavioural observations.

Tableau Story

FOODORDERINGTRENDS / Story1

<https://public.tableau.com/app/profile/mehak.chugh/viz/FOODORDERINGTRENDS/Story1>

7. Performance Testing

7.1. Utilization of Data Filters

Filters and categorical selections were used to support interactive exploration of the dashboard. Their purpose is to allow users to focus the analysis on relevant customer, order, time, weather and behavioural categories without rebuilding the underlying visualizations.

7.2. Number of Calculation Fields

Calculated fields were used where analytical transformations were required, particularly for creating meaningful categories and derived measures used in the dashboard. Calculated logic included segmentation such as age groups and rating categories. The exact workbook-level count should be taken from the final Tableau workbook if an audit count is required.

7.3. Number of Visualizations

The final dashboard uses 9 core visualizations. The project also contains additional business-question analysis, including the meal-type/day-type comparison of delivery fee and time taken to order and the variation of meal type with mood.

8. Conclusion / Observation

The project demonstrates how Tableau can convert food-ordering records into a structured behavioural analysis. The visual framework enables comparison of order value across cities and time periods,

exploration of cuisine and meal-type preferences, and examination of customer ratings, repeat ordering and contextual factors such as weather, discounts and mood.

A key strength of the project is the combination of customer-level and order-level dimensions. Rather than evaluating spending in isolation, the dashboard connects spending with hunger level, age group, cuisine, ratings, weather, discounts, mood and operational measures such as delivery fee and ordering time. This creates a more complete view of food-ordering behaviour.

Overall, the project meets its analytical objective of presenting a clear, interactive Tableau-based view of food ordering behaviour and provides a foundation for further customer segmentation and operational analysis.

9. Future Scope

- Add time-series analysis to identify changes in ordering behaviour over longer periods.
- Develop predictive models for order value, repeat orders or customer preferences.
- Create deeper customer segmentation using clustering or RFM-style approaches.
- Add geographic drill-downs for city, area or restaurant-level analysis.
- Track dashboard KPIs through scheduled refreshes when live or regularly updated data becomes available.
- Integrate additional operational variables such as preparation time, delivery distance and restaurant performance.

10. Appendix

10.1. Source Code (if any)

No separate source-code component is required for the Tableau dashboard. The primary analytical implementation is contained in the Tableau workbook, including calculated fields, filters, worksheets, dashboards and story components.

10.2. GitHub & Project Demo Link

The GitHub repository and project demonstration link should be inserted here when the final repository/demo URL is available. The Tableau dashboard and story links are provided below for direct access to the visualization work.

Dashboard: https://public.tableau.com/app/profile/mehak.chugh/viz/CUSTOMER_ORDERING_ANALYSIS/Dashboard1

Story: <https://public.tableau.com/app/profile/mehak.chugh/viz/FOODORDERINGTRENDS/Story1>

End of Report